

Ex. 41. If $*$ is an operation such that $x * y = 3x + 2y$, find the value of $2 * 3 + 3 * 4$.

Sol. $2 * 3 + 3 * 4 = (3 \times 2 + 2 \times 3) + (3 \times 3 + 2 \times 4) = (6 + 6) + (9 + 8)$
 $= 12 + 17 = 29$.

Ex. 42. If $a^2 + b^2 = 117$ and $ab = 54$, then find the value of $\frac{a+b}{a-b}$.

Sol. $(a+b)^2 = a^2 + b^2 + 2ab = 117 + 2 \times 54 = 225 \Rightarrow a+b = 15$.
 $(a-b)^2 = a^2 + b^2 - 2ab = 117 - 2 \times 54 = 9 \Rightarrow a-b = 3$.
 $\therefore \frac{a+b}{a-b} = \frac{15}{3} = 5$.

Ex. 43. Find the value of $\left(\frac{75983 \times 75983 - 45983 \times 45983}{30000} \right)$.

Sol. Given expression $= \frac{(75983)^2 - (45983)^2}{(75983 - 45983)} = \frac{(a^2 - b^2)}{(a-b)}$, where $a = 75983, b = 45983$
 $= \frac{(a+b)(a-b)}{(a-b)} = (a+b) = (75983 + 45983) = 121966$.

Ex. 44. Find the value of $\left(\frac{343 \times 343 \times 343 - 113 \times 113 \times 113}{343 \times 343 + 343 \times 113 + 113 \times 113} \right)$.

(B.Ed. Entrance, 2010)

Sol. Given expression $= \frac{(a^3 - b^3)}{(a^2 + ab + b^2)}$, where $a = 343, b = 113$
 $= (a-b) = (343 - 113) = 230$.

Ex. 45. If $(x-4)^3 + (x-9)^3 + (x-8)^3 = 3(x-4)(x-9)(x-8)$, find the value of x .

Sol. If $a^3 + b^3 + c^3 = 3abc$, then $a + b + c = 0$.

So, we have: $(x-4) + (x-9) + (x-8) = 0 \Rightarrow 3x - 21 = 0 \Rightarrow 3x = 21 \Rightarrow x = 7$.

Ex. 46. There are certain number of benches in a classroom. If four students sit on each bench then three benches remain unoccupied. If, however, three students sit on each bench then three students remain standing in the class. Find the number of students in the class.

(P.C.S., 2009)

Sol. Let the total number of benches in the class be x .

Case I. When 4 students sit on each bench

In this case, total number of students $= 4(x-3)$.

Case II. When 3 students sit on each bench

In this case, total number of students $= 3x + 3$.

$\therefore 4(x-3) = 3x + 3 \Leftrightarrow 4x - 12 = 3x + 3 \Leftrightarrow x = 15$.

Hence, number of students in the class $= 4(x-3) = 4 \times 12 = 48$.

Ex. 47. A man divides ₹ 8600 among 5 sons, 4 daughters and 2 nephews. If each daughter receives four times as much as each nephew, and each son receives five times as much as each nephew, how much does each daughter receive?

Sol. Let the share of each nephew be ₹ x .

Then, share of each daughter = ₹ $(4x)$; share of each son = ₹ $(5x)$.

So, $5 \times 5x + 4 \times 4x + 2 \times x = 8600 \Leftrightarrow 25x + 16x + 2x = 8600$

$\Leftrightarrow 43x = 8600 \Leftrightarrow x = 200$.

$\therefore$ Share of each daughter = ₹ $(4 \times 200) = ₹ 800$.

Ex. 48. ₹ 6500 were divided equally among a certain number of persons. Had there been 15 more persons, each would have got ₹ 30 less. Find the original number of persons.

(M.A.T., 2005)

Sol. Let the original number of persons be x .

Then, $\frac{6500}{x} - \frac{6500}{(x+15)} = 30 \Leftrightarrow 6500 \left[\frac{x+15-x}{x(x+15)} \right] = 30 \Leftrightarrow 30x(x+15) = 6500 \times 15$